

① Define cloud computing and Explain :

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Definition :

Cloud computing is the on-demand delivery of computing services - including servers, storage database networking software analytics. And intelligence - over the Internet with pay as - you go pricing it allows users to access and uses it resources without owning or maintaining physical infrastructure.

Four Key characteristics :

1) On demand self-services :

User can provision computing resources automatically without human interaction with service providers.

2) Broad Network Access :

Resources are available over the network and accessed through standard mechanism web browsers, mobile, app.

3) Resource Pooling:

Providers, computing, resources, are pooled to serve multiple customers using a multiple tenet model.

4) Rapid Elasticity:

- * Resources can be scaled up or down quickly often automatically.

- * To the consumer available resource after unlimited and can be provided in any quantity at any time.

5) Measured Services:

cloud systems automatically control, and optimize resource usage through metering usage is maintained controlled and reported (pay as you go model).

2) cloud computing has evolved through several stages of technological development.

1) Hardware Evolution:

Early computers were large, expensive, and used by only one user at a time. Later, improvement in processing power, memory, storage, and networking made computers more powerful and affordable.

2) Mainframe and Time-Sharing Systems:

It allows multiple users to access the same computer through time-sharing. This introduced the idea of sharing computer resources among many users.

3) Personal computer & client-server computing:

The development of personal computers led to client-server architecture, where clients requested services and data from centralized servers.

4) Virtualization:

It allows one physical server to run multiple virtual machines. This improved resource utilization, reduced costs, and enabled flexible allocation of computing resources.

5) Internet and Web Evolution:

The growth of the Internet and web 2.0 enabled applications and data to be accessed from anywhere. Web services and APIs allowed different software systems to communicate over the Internet.

6) Modern Cloud Computing:

The combination of high-speed Internet, virtualization, distributed data centers, automation, and web technologies enabled modern cloud platforms.

③ Role of Server Virtualization in cloud computing:

Server virtualization is a technology that divides a single physical server into multiple isolated virtual machines (Vms). Each running of its own, operating system and application.

How it work:

Physical server

1- Hypervision (Vmware, Xen, KVM)

1-Vm₁ (Windows, App A)

Vm₂ (Linux, App B)

Vm₃ (Linux, App C)

Nm₄ (Windows, App C)

④ Role of Data centers in Delivering cloud Services :

A Data centre is a physical facility that houses, computer system storage, system networking, equipment and related infrastructure for storing, processing and managing data.

Component	Function
Servers	Computing Power
Storage	Data storage.
Networking	Switches, routers
Power Systems	UPS, generators, power distribution.
Cooling	AC units, cooling system.
Security	Physical and cybersecurity measures.

Role in cloud Services:

- 1) Physical Foundation
- 2) Redundancy and Reliability
- 3) Global reach
- 4) Economics of scale
- 5) Security.

⑤ Differentiate IaaS, PaaS, SaaS, XaaS

Real world Example:

1) IaaS - AWS:

- * Ex: Launching a virtual server on AWS EC2

- * User gets: virtual machine CPU, RAM

- * User configuration: Operating system, security application.

- * Provider manages: Physical hardware.

2) PaaS - Google App Engine:

- * Example: Deploying a python / Java

- * User gets: Development platform and runtime

- * User configures: Application code only.

- * User manages: only applications and data

- * Provider manages: OS, runtime.

3) SaaS . Salesforce CRM :

- * Ex : Using Salesforce . Pay customer .
- * User gets : Complete CRM application
- * User configures : Settings
- * User manages : Nothing technical
- * Provides : Everything.

4) XaaS :

- * Example : Flight Reservation
- * User gets : Cloud based application flat
- * Service : Custom application
- * Provider manages : Everything while user.
- * Policies : on building.